

ENTERPRISE CLOUD SOLUTION

Complete Cloud Implementation Project

Professional Portfolio Documentation

Project Enterprise Cloud Migration	Version v2.0 — June 2026
Classification Confidential — Internal Use	Status In Progress — Phase 2

1. Executive Summary

2. Cloud Architecture Design

2.1 Architecture Principles

The architecture is guided by the AWS Well-Architected Framework, extended with Microsoft Azure's Cloud Adoption Framework for hybrid scenarios. Six foundational pillars govern all design decisions:

- 1. Operational Excellence — Automated operations, continuous improvement loops, runbook-as-code
- 2. Security — Defense-in-depth, least privilege, encryption at rest and in transit
- 3. Reliability — Multi-AZ deployments, chaos engineering, automated failover
- 4. Performance Efficiency — Right-sizing, auto-scaling, cache optimization
- 5. Cost Optimization — Reserved instances, spot workloads, FinOps governance
- 6. Sustainability — Green regions, efficient packaging, compute density optimization

2.2 Network Topology

The network design implements a hub-and-spoke model with a Transit Gateway connecting all VPCs and on-premises environments through dedicated AWS Direct Connect (10 Gbps) and Azure ExpressRoute (5 Gbps) circuits.

Layer	Component	Technology	Purpose
Ingress	Global Load Balancer	AWS CloudFront + WAF	DDoS protection, SSL termination, geo-routing
DMZ	API Gateway	Kong Enterprise	Rate limiting, auth, API versioning
Application	Service Mesh	Istio + Envoy	mTLS, traffic management, observability
Data	Private Subnets	VPC Peering + PrivateLink	Isolated data plane, no internet exposure
Management	Bastion / ZTNA	Tailscale Enterprise	Zero-trust remote access, MFA enforced
Edge	On-Prem	VMware + NSX	Legacy workloads, low-latency processing

2.3 Compute Strategy

2.3.1 Container Platform

Amazon EKS (managed Kubernetes) serves as the primary compute platform, supplemented by Azure AKS for disaster recovery. Cluster configuration follows a node-pool segmentation strategy:

Node Pool	Instance Type	Scaling	Workload Type
system	m6i.2xlarge	Fixed (3 nodes)	Kubernetes system components
general	m6i.4xlarge	2–20 nodes (HPA/KEDA)	Stateless microservices
memory	r6i.4xlarge	1–10 nodes	In-memory caches, JVM workloads
gpu	g4dn.xlarge	0–5 nodes (scale-to-zero)	ML inference, image processing
spot	Diversified (m/c/r families)	0–50 nodes	Batch, CI/CD runners, dev

2.3.2 Serverless Integration

- AWS Lambda for event-driven processing (S3 triggers, SNS/SQS consumers)
- AWS Fargate for scheduled batch jobs and short-lived container tasks
- Azure Functions for cross-cloud integration workflows and ETL pipelines

2.4 Data Architecture

Data is classified into four tiers with distinct storage, access, and retention policies:

Tier	Classification	Storage Service	Retention	Encryption
Hot	Operational (< 90 days)	Amazon RDS (Multi-AZ), ElastiCache	90 days	AES-256 + CMK
Warm	Analytics (90d–2yr)	Amazon Redshift, S3 Intelligent-Tiering	2 years	AES-256 + CMK
Cold	Archive (2yr–7yr)	S3 Glacier Instant Retrieval	7 years	AES-256 + CMK
Frozen	Compliance (7yr+)	S3 Glacier Deep Archive	10 years	AES-256 + CMK

2.5 High Availability & Disaster Recovery

RTO / RPO Target	Strategy	Region	Automation
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RTO < 15 min / RPO < 5 min	Active-Active	us-east-1 + us-west-2	Route 53 health checks + Lambda failover
RTO < 1 hr / RPO < 1 hr	Active-Passive (warm)	Azure East US 2	Azure Traffic Manager + runbooks
RTO < 4 hr / RPO < 24 hr	Backup-Restore	AWS ap-southeast-1	Automated AMI/snapshot replication

3. Security & Compliance Framework

3.1 Security Architecture

The security model adopts NIST SP 800-207 Zero Trust Architecture principles, eliminating implicit trust and enforcing continuous verification for all users, devices, and workloads regardless of network location.

3.2 Identity & Access Management

- Centralized IdP: Okta Enterprise with SAML 2.0 / OIDC federation to AWS IAM Identity Center and Azure Entra ID
- Privileged Access Management: CyberArk PAM for just-in-time elevation with session recording
- Service Accounts: IRSA (IAM Roles for Service Accounts) and Workload Identity for pods — no long-lived credentials
- MFA enforcement: FIDO2/WebAuthn (phishing-resistant) for all human accounts
- Secrets Management: HashiCorp Vault Enterprise with auto-rotation and dynamic credentials

3.3 Compliance Mapping

Requirement	Standard	Control Implementation	Audit Frequency
Data Encryption	GDPR Art. 32 / ISO 27001 A.10	AES-256 at rest, TLS 1.3 in transit	Continuous (AWS Config)
Access Logging	SOC 2 CC6.3 / PCI DSS 10	CloudTrail + VPC Flow Logs centralized in S3	Real-time SIEM
Vulnerability Mgmt	ISO 27001 A.12.6 / SOC 2 CC7	Amazon Inspector + Snyk in CI/CD pipeline	Weekly scans
Incident Response	GDPR Art. 33 / SOC 2 CC7.4	Automated IR playbooks in AWS Security Hub	Quarterly drills
Data Residency	GDPR / Local regulations	Data classification tags + S3 bucket policies	Monthly review
Penetration Testing	PCI DSS 11.3 / SOC 2	Quarterly external + annual red team exercise	Per schedule

3.4 Threat Detection & Response

- Amazon GuardDuty for ML-based threat detection across CloudTrail, DNS, and VPC Flow Logs
 - Microsoft Sentinel as SIEM/SOAR hub with 45-day hot retention and 12-month cold
 - AWS Security Hub aggregating findings from 15+ security services with ASFF normalization
 - Custom detection rules leveraging MITRE ATT&CK framework coverage mapping
 - Mean time to detect target: < 5 minutes; mean time to respond: < 30 minutes
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4. Implementation Roadmap

4.1 Phased Delivery Plan

The migration follows a five-phase model spanning 18 months, with each phase delivering independent business value and minimizing risk through incremental transition.

Phase	Timeline	Scope	Key Deliverables	Success Criteria
0 — Foundation	Weeks 1–6	Landing zone, networking, IAM	AWS Control Tower, Hub VPC, Okta federation	Zero findings in CIS benchmark scan
1 — Platform	Weeks 7–16	EKS clusters, CI/CD, observability	3 EKS clusters, GitOps pipeline, Grafana stack	< 15 min deployment, 100% observability coverage
2 — Migration Wave 1	Weeks 17–28	Non-critical workloads (30% of apps)	10 microservices containerized & deployed	All SLAs met, zero data loss
3 — Migration Wave 2	Weeks 29–44	Business-critical workloads (60% of apps)	Core platform migrated, DR validated	99.99% availability demonstrated
4 — Optimization	Weeks 45–52	FinOps, performance tuning, automation	Cost dashboards, auto-scaling policies	35% cost reduction achieved
5 — Operationalize	Weeks 53–72	Remaining workloads, legacy decom	Full cloud migration, DC exit completed	All KPIs met, data center decommissioned

4.2 CI/CD Pipeline Architecture

The DevSecOps pipeline implements a shift-left security model with automated quality gates at each stage:

- Developer commits code → GitHub Enterprise (branch protection, signed commits required)
- Pre-commit hooks run: Gitleaks (secret scanning), ESLint/Black (linting), unit tests
- Pull Request triggers: SAST (Semgrep), dependency scan (Snyk), code review (2 approvals)
- Merge to main: Integration tests, container image build (Kaniko), Trivy vulnerability scan
- Image signing (Cosign + Sigstore) and push to Amazon ECR with immutable tags
- ArgoCD GitOps sync to dev → staging → production (manual gate for production)
- Post-deployment: DAST (OWASP ZAP), smoke tests, SLO verification, rollback on failure

4.3 Infrastructure as Code Standards

All infrastructure is defined, versioned, and deployed through IaC, eliminating manual configuration drift:

- Terraform (HashiCorp) for cloud infrastructure — modules published to internal registry
- Helm charts for Kubernetes application packaging — stored in OCI-compatible registry
- Crossplane for Kubernetes-native cloud resource provisioning
- Ansible for OS-level configuration and compliance hardening (CIS benchmarks)
- OPA / Conftest for pre-deployment policy validation in CI pipeline
- Checkov for Terraform security scanning — blocks >MEDIUM severity findings

5. Cost Management & FinOps

5.1 Total Cost of Ownership Analysis

Cost Category	Year 1	Year 2	Year 3	3-Year Total
AWS Compute (Reserved + Spot)	\$480,000	\$420,000	\$380,000	\$1,280,000
AWS Storage & Database	\$180,000	\$160,000	\$145,000	\$485,000

Azure DR Environment	\$120,000	\$110,000	\$100,000	\$330,000
Network & Transfer	\$85,000	\$78,000	\$72,000	\$235,000
Security & Compliance Tools	\$95,000	\$90,000	\$85,000	\$270,000
Migration & Professional Services	\$320,000	\$80,000	\$40,000	\$440,000
Training & Certification	\$45,000	\$30,000	\$25,000	\$100,000
TOTAL CLOUD SPEND	\$1,325,000	\$968,000	\$847,000	\$3,140,000
Legacy Infrastructure Cost (avoided)	(\$1,200,000)	(\$1,400,000)	(\$1,600,000)	(\$4,200,000)
NET SAVINGS	(\$125,000)	\$432,000	\$753,000	\$1,060,000

5.2 FinOps Governance

- Cloud cost tagging taxonomy enforced via AWS Config rules — 100% resource tagging mandatory
- Monthly FinOps review cadence with engineering, finance, and product stakeholders
- Savings Plans and Reserved Instance coverage target: 70% of baseline compute
- Spot instance utilization target: 40% of non-critical batch and development workloads
- Right-sizing recommendations automated via AWS Compute Optimizer with Slack alerts
- Budget alerts at 80% / 100% / 120% of monthly forecast with escalation runbooks

6. Operations & Governance

6.1 Observability Stack

Signal Type	Collection	Storage	Visualization	Alerting
Metrics	Prometheus + OTLP	Thanos (S3 long-term)	Grafana	Alertmanager + PagerDuty
Logs	Fluent Bit	OpenSearch (Serverless)	Grafana / Kibana	Elastalert2
Traces	OTLP (OpenTelemetry)	AWS X-Ray / Tempo	Grafana Tempo	Trace-based alerts
Events	AWS CloudTrail / K8s Audit	S3 + Athena	Grafana	AWS EventBridge
Synthetics	k6 / Playwright	InfluxDB	Grafana	PagerDuty on SLO breach

6.2 SRE & SLO Framework

Service Level Objectives are defined per service tier, with error budgets tracked in real-time:

Service Tier	Availability SLO	Latency SLO (p99)	Error Budget (30d)	Burn Rate Alert
Tier 1 (Revenue-critical)	99.99%	< 200ms	4.3 minutes	1-hour burn > 2% → page
Tier 2 (Business-critical)	99.95%	< 500ms	21.6 minutes	6-hour burn > 5% → ticket
Tier 3 (Internal tools)	99.5%	< 2 seconds	3.6 hours	24-hour burn > 10% → email

6.3 Incident Management

- On-call rotation: 24x7 coverage with primary + secondary responders per tier
 - Incident severity classification: P0 (all-hands), P1 (on-call), P2 (next business day), P3 (backlog)
 - Post-incident reviews mandatory for P0/P1 — blameless, published within 5 business days
 - Runbooks stored as code in GitHub, linked from PagerDuty alerts for guided response
 - Chaos engineering: monthly GameDay exercises using AWS Fault Injection Simulator
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6.4 Team Structure & RACI

Activity	Cloud Arch	Platform Eng	Dev Teams	SecOps	FinOps
Architecture decisions	R/A	C	I	C	I
IaC development	A	R	C	I	I
Application deployment	I	A	R	I	I
Security policy	C	I	I	R/A	I
Cost optimization	C	C	I	I	R/A
Incident response	C	R	C	A	I

7. Risk Register

Risk ID	Description	Likelihood	Impact	Rating	Mitigation Strategy
R-01	Migration data loss during cutover	Low	Critical	HIGH	Blue-green deployments, dual-write phase, automated validation suite
R-02	Vendor lock-in to AWS primitives	Medium	High	HIGH	Cloud-agnostic abstractions (K8s, OTLP, Terraform), documented portability plan
R-03	Skill gap — team cloud maturity	High	Medium	HIGH	AWS/Azure cert program, embedded cloud engineers, CoE model
R-04	Cost overrun in Year 1	Medium	Medium	MEDIUM	FinOps guardrails, budget alerts, bi-weekly spend reviews
R-05	Security breach during migration	Low	Critical	MEDIUM	Dedicated security workstream, penetration testing pre-cutover
R-06	Legacy dependency undiscovered	High	Medium	MEDIUM	Application dependency mapping (AWS Migration Hub), extended parallel-run
R-07	Regulatory non-compliance	Low	High	MEDIUM	Compliance-as-code from day 1, quarterly external audits
R-08	DR failover untested	Medium	High	MEDIUM	Mandatory quarterly DR exercises, automated failover runbooks

Appendix A — Glossary

Term	Definition
AKS	Azure Kubernetes Service — managed Kubernetes on Microsoft Azure
ArgoCD	GitOps continuous delivery tool for Kubernetes
CMK	Customer Managed Key — encryption keys controlled by the organization
DAST	Dynamic Application Security Testing — runtime vulnerability scanning
EKS	Elastic Kubernetes Service — managed Kubernetes on AWS
FinOps	Financial Operations — cloud cost management practice
GitOps	Git as the single source of truth for infrastructure and application state
IaC	Infrastructure as Code — defining infrastructure in version-controlled files
IRSA	IAM Roles for Service Accounts — pod-level AWS identity in EKS
mTLS	Mutual TLS — bidirectional certificate authentication between services
OPA	Open Policy Agent — policy enforcement engine
OTLP	OpenTelemetry Protocol — vendor-neutral observability data format
RPO	Recovery Point Objective — maximum acceptable data loss window
RTO	Recovery Time Objective — maximum acceptable downtime
SAST	Static Application Security Testing — code-level vulnerability scanning
SIEM	Security Information & Event Management — centralized log analysis
SLO	Service Level Objective — internal reliability target
SOAR	Security Orchestration, Automation and Response
ZTNA	Zero Trust Network Access — identity-based network access control

Appendix B — Reference Architecture

The following cloud provider documentation and frameworks informed this architecture:

- AWS Well-Architected Framework — aws.amazon.com/architecture/well-architected
- Microsoft Azure Cloud Adoption Framework — learn.microsoft.com/azure/cloud-adoption-framework
- NIST SP 800-207 Zero Trust Architecture — csrc.nist.gov
- CIS Benchmarks for AWS and Kubernetes — cisecurity.org
- CNCF Cloud Native Landscape — landscape.cncf.io
- HashiCorp Terraform Best Practices — developer.hashicorp.com

Appendix C — Document Control

Version	Date	Author	Changes
1.0	March 2026	Cloud Architecture Team	Initial draft — foundation and Phase 0 scope
1.5	May 2026	Cloud Architecture Team	Added Phase 1–3 detail, FinOps section, risk register
2.0	June 2026	Cloud Architecture Team	Full document — all phases, compliance mapping, SRE framework